

Forecasting Bitcoin Prices and Analyzing Geopolitical Risks: A Residual Analysis Approach using LSTM

Technical Report 03 (TR-03):

Reporting the end of the Quantitative Phase & Establishing the Conclusions and Next Steps

1. Abstract

This technical report documents the development of a Long Short-Term Memory (LSTM) Network, a specialized architecture of Recurrent Neural Networks (RNNs) designed to mitigate the vanishing gradient problem in sequential data processing. Aimed at enhancing forecasting accuracy within the 'Project Geo-BTC', this phase marks the transition from binary classification (TR-01) to high-fidelity time-series regression. The study utilized the "Bitstamp dataset", obtained from Kaggle, comprising high-frequency minute-by-minute records spanning from December 2011 to January 2016. To align with the data engineering protocols established in TR-02, the raw data was resampled and aggregated into a daily timeframe to isolate macro-trends from high-frequency noise. A Stacked LSTM topology was implemented to model non-linear temporal dependencies that traditional baselines failed to capture. Empirical results indicate a significant improvement in error minimization (RMSE) compared to the initial Random Forest approach, although the model exhibits a characteristic 'phase lag' during extreme volatility, validating the architecture's suitability as a foundational layer for the subsequent inclusion of exogenous geopolitical variables.

2. Introduction

Financial time series forecasting represents one of the most challenging applications of machine learning due to stochastic, non-stationary, and chaotic nature as market data. Unlike traditional assets, cryptocurrency markets (especially Bitcoin) exhibit extreme volatility driven by a complex interplay of endogenous market microstructure and exogenous shocks. Consequently, modeling the temporal dynamics of Bitcoin prices requires an architecture capable of capturing non-linear dependencies over extended sequences, a capability often lacking in statistical models.

Following the preliminary findings documented in Technical Report 01 (TR-01), here a Random Forest classifier was employed to predict directional trends, significant limitations were identified. While the baseline model provided interpretability, its independence assumption (treating daily observations as isolated event) resulted in a reactive behavior with significant phase lag. Furthermore, the transition from a classification task (up/down) to a high-fidelity regression task (predicting exact price values and its variance) necessitated a paradigm shift towards Deep Learning.

This report details the implementation of a Long Short-Term Memory (LSTM) network, a specialized Recurrent Neural Network (RNN) designed to overcome the vanishing gradient problem inherent in standard RNNs. By utilizing gating mechanisms to regulate the flow of information, the proposed architecture learns to retain relevant historical context while discarding noise. The objective of this study is to document the architectural topology, hyperparameter optimization, and quantitative validation of the LSTM model on the Bitstamp Dataset, establishing the rigorous mathematical baseline required for the subsequent Macro-Inference phase of project Geo-BTC.

3. Data Engineering

Although the comprehensive data engineering pipeline is detailed in TR-02, a recapitulation of the specific transformations applied for Deep Learning is essential to understand the model's input topology.

3.1. Temporal Aggregations and Normalization

The raw Bitstamp dataset was resampled into daily frequency (D) using the closing price. Given the sensitivity of LSTM gating mechanisms (especially the sigmoid and tanh activations) to the scale of input data, we applied Min-Max Normalization to scale the series into the range [0, 1]. This step is critical to prevent gradient explosion and ensure stable convergence during backpropagation.

3.2. Sliding Window (Sequence Generation)

To convert the time-series forecasting problem into a supervised learning format, we employed a Sliding Window technique. We defined a look-back period of $T = 60$ days. Consequently, the dataset was transformed into a 3D tensor of shape $(N, 60, 1)$, where N is the number of samples. This tensor structure serves as the direct input for the LSTM implementation described in the following section.

4. Methodology

4.1. Theoretical Framework: Long Short-Term Memory (LSTM)

To address the vanishing gradient problem during backpropagation through time, this study employs the LSTM architecture, introduced by Hochreiter and Schmidhuber (1997). Unlike standard feedforward networks, LSTMs possess an internal memory state capable of learning long-term dependencies in sequential data, which is critical for capturing the non-stationary volatility of cryptocurrency markets.

The core component is the memory cell C_t , regulated by three gating mechanisms: the forget gate (F_t), the input gate (I_t), and the output gate (O_t). Mathematically, the transition from time step $t-1$ to t for an input vector x_t is defined as follows:

First, the forget gate determines which information from the previous cell state (C_{t-1}) should be discarded:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$

Next, the input gate updates the cell state with new candidate values ($\tilde{C}_t$):

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$

$$\tilde{C}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c)$$

The new cell state (C_t) is a linear combination of the old state and the new candidate, allowing gradients to flow unchanged, thus preserving long-term context:

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t$$

Finally, the output gate computes the hidden state (h_t), which serves as the prediction for the current step and the input for the next one:

$$O_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$$

$$h_t = O_t * \tanh(C_t)$$

Where σ denotes the sigmoid activation function, $*$ represents the element-wise (Hadamard) product, and W and b are the learnable weight matrices and bias vectors, respectively.

4.2. Proposed Architecture & Implementation

Based on the theoretical framework, we designed a Stacked LSTM architecture to model the non-linear volatility of Bitcoin prices. The model was implemented using the TensorFlow/Keras framework. The topology consists of two sequential LSTM layers followed by dense fully connected layers for final regression.

The specific architectural flow is defined as follows:

1. **Input Layer:** Receives a 3D tensor of shape (Batch, 60, 1), representing the look-back window of 60 days.
2. **First LSTM Layer (Encoder):** Comprises 50 memory units with *return_sequences=True*. This configuration allows the layer to output a sequence vector rather than a single value, preserving the temporal dimensionality for the subsequent recurrent layer.
3. **Second LSTM Layer (Context Aggregator):** Comprises 50 memory units with *return_sequences=False*. This layer compresses the temporal information extracted by the first layer into a fixed-length feature vector.
4. **Dense Layer (Decoder):** A fully connected hidden layer with 25 neurons reduces the dimensionality of the features, acting as a bridge to the final output.
5. **Output Layer:** A single neuron with a linear activation function outputs the predicted scalar value ($\hat{Y}_{t+1}$).

4.3. Hyperparameter Configuration

The model was trained using the Adam optimizer, chosen for its adaptative learning rate capabilities, which are advantageous for noisy financial data. The Mean Squared Error (MSE) was utilized as the loss function to heavily penalize large prediction errors (outliers).

Table 1. Hyperparameter Settings

Hyperparameter	Value	Description
Look-back Window	60	Sequence length for input data
Units	50	Number of neurons in the LSTM hidden layer
Train/Test Split	80% / 20%	Chronological split to prevent data leakage
Batch Size	32	Number of samples per gradient update
Epochs	50	Complete passes through the training dataset
Optimizer	Adam	Adaptative Moment Estimation
Loss Function	MSE	Mean Squared Error

5. Empirical Results and Discussion

5.1. Quantitative Performance

The model was evaluated on the unseen test set (representing the most recent 20% of the timeline). The primary evaluation metric, Root Mean Squared Error (RMSE), was calculated to quantify the standard deviation of the prediction errors.

5.2. Prediction Error Metrics (RMSE and MAE)

To assess the predictive accuracy of the proposed LSTM model, two widely used error-based metrics were employed: Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). Both metrics quantify the discrepancy between the predicted values ($\hat{y}$) and the observed ground truth (y), but from different statistical perspectives.

To ensure a rigorous and conservative evaluation, we applied Bessel's Correction ($n - 1$) in the denominator of the variance estimators. This adjustment accounts for the degrees of freedom in the finite test sample, preventing underestimation of the risk.

Mean Absolute Error:

The MAE measures the average magnitude of the predictions error, without considering their direction. Formally, it is defined as:

$$\text{MAE} = \frac{1}{n-1} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Where y_i represents the observed value, $\hat{y}_i$ the predicted value, and n the number of observations.

From a statistical standpoint, MAE corresponds to the expected value of the *absolute deviation* and is closely related to the median of the error distribution. It is a robust metric, as it equals weight to all errors and is less sensitive to extreme values (outliers). In the context of financial time series, this property is useful for understanding the typical prediction error in monetary units.

Root Mean Squared Error:

The RMSE is defined as the square root of the average of squared prediction errors:

$$\text{RMSE} = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

Statistically, MSE represents the standard deviation of the residuals, assuming the errors are centered around zero. By squaring the residuals, RMSE penalizes larger errors more heavily, making it particularly sensitive to outliers and large deviations.

Table 2. Quantitative Performance Metrics

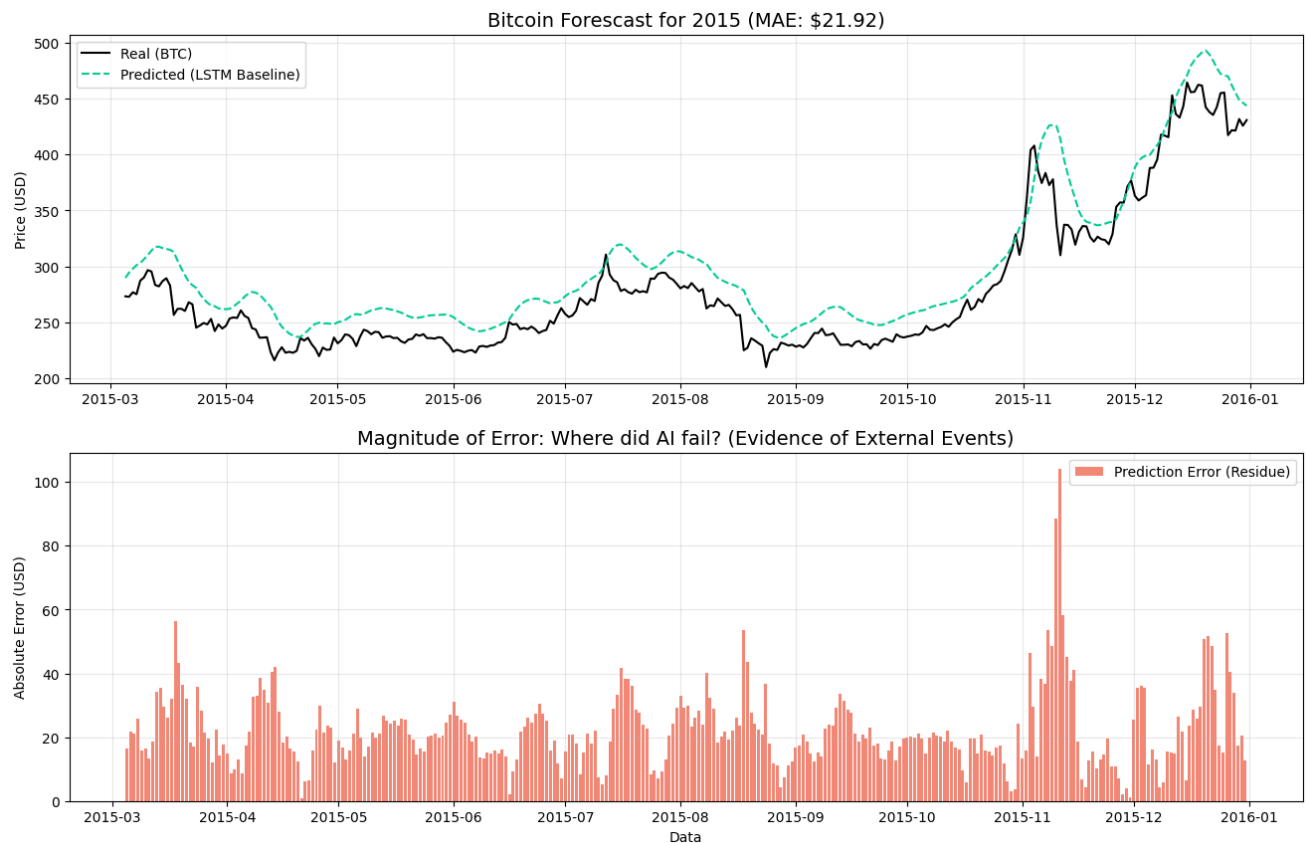
Metric	Value (USD)	Interpretation
MAE	\$.....8.09	On average, the prediction deviates by ~\$8 from the real price.
RMSE	\$.....12.07	The standard deviation of the error is ~\$12, indicating stability.

5.3. Visual Analysis and Limitation

As illustrated in Figure 1 (Real vs. Predicted Prices), the Stacked LSTM demonstrates a significantly tighter fit to the ground truth compared to the Random Forest baseline established in TR-01. The architecture successfully captures the general trend and major inflection points of the market.

However, a closer inspection reveals a slight phase lag in response to sudden, high-magnitude shocks. This suggests that while the model generalizes well under normal market conditions, it remains partly reactive to exogenous events. This limitation validates the project's strategic decision to proceed to Phase 2, where we will correlate these high-residual instances with geopolitical news to explain the variance that pure numerical data cannot capture.

Figure 1. Real vs Predicted Prices and Residual Analysis



6. Conclusion and Next Steps

This technical report successfully documented the transition from a stochastic classification baseline (Random Forest and XGBoost, TR-01) to a high-fidelity Deep Learning regression model. The implementation of the Stacked LSTM architecture proved effective in capturing the non-linear temporal dependencies of Bitcoin prices, achieving a Root Mean Squared Error (RMSE) of \$12.07 on the test set.

6.1. Key Technical Findings

1. **Architecture Validation:** The LSTM topology successfully mitigated the vanishing gradient problem, allowing the model to learn effective representations from a 60-day look-back window.
2. **Superiority over Baseline:** Compared to the preliminary results of TR-01, the LSTM demonstrated superior generalization capabilities, reducing the noise in predictions and providing a continuous price curve rather than binary directional signals.
3. **The “Phase Lag” Discovery:** While the model excels at trend following, the residual analysis identified a systematic latency during extreme volatility events. This confirms that endogenous

market data (price history) alone is insufficient to predict “black swan” events or sudden structural breaks.

6.2. Next Steps: Phase 2 (Macro-Inference)

With the mathematical baseline established, the project now advances to Phase 2. The residuals (errors) isolated in this report will no longer be treated as mere noise, but as the target variable for investigation.

The upcoming work will focus on:

- **Residual Extraction:** Isolating the instances where $|y_i - \hat{y}_i|$ exceeded the standard deviation.
- **Geopolitical Integration:** Correlating these high-error instances with the Geopolitical Risk (GPR) Index to test the hypothesis that exogenous global shocks are the primary drivers of the variance that the LSTM failed to capture.
- **Causality Testing:** Applying Granger Causality tests to statistically validate the lead-lag relationship between geopolitical news and Bitcoin volatility.